Towards Industrial Anomaly Cognition: A Large-Scale Knowledge Benchmark, Challenges, and Baselines

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Abstract—With the development of intelligent manufacturing, industrial anomaly analysis is extending from anomaly detection to anomaly understanding and cognition. However, existing research still faces two limitations: **i)** current benchmarks provide limited category and scene coverage, while anomaly understanding, and cognition remain isolated, lacking a continuous cognition process. **ii)** Existing multi-modal methods make insufficient use of internal and external knowledge. Internal knowledge typically relies on normal image reconstruction and lacks explicit normal-to-anomaly relation modeling, while external knowledge often directly introduces normal images without sufficiently considering their relevance to the query image. To address these limitations, we construct MMIK-400K and propose ADSeeker++. To the best of our knowledge, MMIK-400K is the first large-scale industrial multi-modal benchmark that unifies Scene Knowledge Pretraining, Anomaly Understanding, and OpenQA Cognition. It contains 461,388 real industrial images from 432 object categories, 1,271,155 closed QA pairs, and 133,140 open QA pairs. ADSeeker++ jointly models internal and external knowledge: internal knowledge explicitly learns normal-to-anomaly relations, while external knowledge models the relevance between the query image and normal references. Extensive experiments validate the benchmark value of MMIK-400K and the effectiveness of ADSeeker++ on anomaly understanding and cognition tasks. MMIK-400K and ADSeeker++ jointly advance industrial anomaly research from localized anomaly reasoning toward more systematic anomaly cognition. An earlier version of this work was presented at CVPR 2026. This paper extends it in both benchmark construction and method design. The dataset and code are available at https://github.com/hellozzk/MMIK-400K.

Index Terms—Industrial Anomaly Cognition, Large-Scale Knowledge Benchmark, Visual-Language Baselines.

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1 Introduction

Industrial Anomaly Detection (IAD) has been widely adopted in industrial automation, providing essential support for product quality control and intelligent manufacturing. Early IAD research mainly focused on visual anomaly detection. Benchmarks such as MVTec AD [1], VisA [2], and Real-IAD [3] established standardized evaluation benchmarks for anomaly classification and pixel-level localization, while methods including DNPR [4], FreqPrompt-AD [5], AnomalyLVM [6], and DSSM-MAPL [7] further improved anomaly detection, localization, and cross-category generalization. With the development of vision-language models [8]–[10], the research paradigm has gradually expanded from anomaly detection to anomaly understanding. JUDO [11], AnomalyR1 [12], and AnomalyGPT [13] support anomaly description, and cause explanation, while multi-modal benchmarks such as MMAD [14] and MANTA [15] further extend evaluation to anomaly understanding.

However, existing industrial anomaly research still mainly focuses on isolated closed and open tasks (Fig 1). This makes systematic anomaly cognition evaluation difficult [14], [15]. On the one hand, existing benchmarks are often built by aggregating previous datasets or collecting data in controlled environments. Increasing data scale does not broaden the coverage of object categories and scenes variations. Meanwhile, such as Tab. 2, existing benchmarks typically cover only one stage of Anomaly Understanding and OpenQA Cognition (OpenQA Cognition, which requires models to autonomously explain anomalous and make anomaly detection decisions without predefined candidate answers), lacking a continuous evaluation process across these tasks. On the other hand, existing multi-modal anomaly reasoning methods often combine internal and external knowledge [11], [12]. Internal knowledge mainly focuses on normal patterns while overlooking the relation between normal and anomalous states, making anomaly deviations difficult to model accurately. External knowledge often directly introduces normal images without explicitly modeling their relevance to the query image, which may result in visually mismatched references.

To address the limitations of existing industrial anomaly

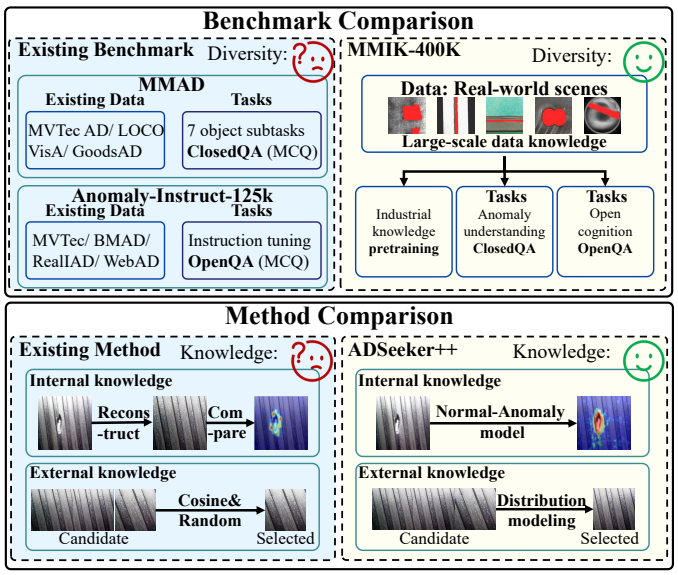

Fig. 1. Comparison of benchmarks and methods. MMIK-400K provides a large-scale corpus of real-world industrial images that unifies scene knowledge pretraining, anomaly understanding with ClosedQA, and open cognition with OpenQA. Compared with reconstruction-based anomaly analysis and reference selection, ADSeeker++ explicitly models normal-to-anomaly relations for internal knowledge and the relevance distribution between the query image and normal references for external knowledge.

benchmarks in category coverage, knowledge learning, and cognition evaluation, we construct the Multi-Modal Industrial Knowledge Benchmark-400K (MMIK-400K). To the best of our knowledge, MMIK-400K is the first large-scale industrial multi-modal benchmark that unifies Scene Knowledge Pretraining, Anomaly Understanding, and OpenQA Cognition. It contains 461,388 real industrial images covering 432 object categories, together with 1,271,155 closed QA pairs and 133,140 open QA pairs. Unlike previous benchmarks that focus on isolated tasks, MMIK-400K establishes a continuous evaluation process, extending industrial anomaly evaluation from isolated task to a unified cognition framework.

To address the insufficient use of internal and external knowledge in existing multi-modal anomaly reasoning, we further propose ADSeeker++. ADSeeker++ introduces two key components: ADAgent and Image-RAG. ADAgent explicitly models normal visual distributions and their normal-to-anomaly relations to learn fine-grained internal anomaly knowledge. Image-RAG models the similarity distribution between the query image and candidate normal images to select more relevant visual references. Finally, the vision-language model integrates these two complementary knowledge sources for fine-grained anomaly cognition.

Part of this work was initially presented at CVPR 2026 [16]. We extend the conference version in three aspects. **1)** Compared with previous benchmark, which mainly focuses on anomaly detection, MMIK-400K expands the scope to a large-scale unified benchmark covering Scene Knowledge Pretraining, Anomaly Understanding, OpenQA Cognition, and visual knowledge augmentation. **2)** Compared with the early version of ADSeeker, the new ADSeeker++ introduces knowledge learning framework to improve anomaly cognition. **3)** We further provide more extensive experiments and deeper analyses to validate the effectiveness of the MMIK-400K and ADSeeker++. Our main contributions are summarized as follows:

- **New Cognition Benchmark:** We construct the first large-scale multi-modal industrial benchmark that unifies Scene Knowledge Pretraining, Anomaly Understanding, and OpenQA Cognition. Unlike previous MulA, MMIK-400K contains 461,388 real industrial images covering 432 object categories, together with 1,271,155 closed QA pairs and 133,140 open QA pairs, enabling unified evaluation from knowledge learning to anomaly cognition.
- **New Knowledge baseline:** We propose ADSeeker++, an internal-external knowledge framework for industrial anomaly cognition. Unlike previous ADSeeker, we design ADAgent models normal visual distributions and their normal-to-anomaly structural relations to learn internal anomaly knowledge, while Image-RAG uses similarity-distribution modeling to select external normal visual knowledge. The two knowledge sources jointly improve fine-grained anomaly understanding and cognition.
- **Comprehensive Evaluation:** We establish an evaluation covering both closed anomaly understanding and open anomaly cognition. Extensive experiments on MMIK-400K, MMAD [14], and other benchmarks validate the benchmark value of MMIK-400K and the effectiveness of ADSeeker++ for anomaly understanding and cognition.

2 Related Work

2.1 Industrial Anomaly Benchmarks

Early industrial anomaly detection benchmarks mainly focused on image-level classification and pixel-level localization under controlled conditions. Datasets such as DAGM [17], KolektorSDD [18], BTAD [19], and MPDD [20] provided standard benchmarks for surface defects and local anomalies, but their category and scene diversity remained limited. MVTec AD [1] later introduced multiple object and texture categories. Subsequent datasets further extended evaluation to logical anomalies, 3D observations, multi-view acquisition, and more diverse industrial objects, including MVTec LOCO-AD [21], MVTec 3D-AD [22], VisA [2], and Real-IAD [3]. However, these benchmarks still cover a limited range of industrial objects and scenarios, making it difficult to fully represent the category and scene diversity of real-world production environments.

Recent multi-modal benchmarks have extended industrial anomaly analysis to anomaly understanding. Benchmarks such as MMAD [14], Anomaly-Instruct [23], MANTA [15], and MMR-AD [24] begin to evaluate the anomaly understanding ability of multi-modal large language models. However, many existing benchmarks are still built from a limited set of existing datasets or controlled environments, restricting the coverage of object categories and production environments. Meanwhile, anomaly reasoning is usually evaluated as separate task (Tab. 2). In contrast, MMIK-400K unifies Scene Knowledge Pretraining,

Anomaly Understanding, and OpenQA Cognition, while further introducing category-related normal visual knowledge. MMIK-400K extends industrial anomaly evaluation from isolated tasks to a continuous process covering knowledge learning, anomaly understanding, and cognition.

2.2 Industrial Anomaly Detection

Industrial anomaly detection methods can be broadly categorized into embedding-based, reconstruction-based, and anomaly-synthesis approaches. Embedding-based methods model normal features and detect anomalies according to their deviations from the normal distribution. SPADE [25], PaDiM [26], and PatchCore [27] mainly rely on normal feature matching or distribution modeling. FastFlow [28], CFLOW-AD [29], SimpleNet [30], EfficientAD [31], AnomalyDINO [32], UniVAD [33], and DNPR [4] further improve feature modeling, detection efficiency, and cross-category generalization. Reconstruction-based methods identify anomalies through image or feature reconstruction errors, including MAFR [34], Reverse Distillation [35], and Dinomaly [36]. Anomaly-synthesis methods learn anomaly representations by generating pseudo-anomalous samples, with CutPaste [37] and DRAEM [38] as representative examples. These methods have advanced anomaly classification, pixel-level localization, and cross-category generalization, but still mainly rely on visual information for anomaly detection. Vision-language models have further extended industrial anomaly from visual detection to semantic understanding. WinCLIP [39] performs anomaly recognition through image-text matching, while AnomalyCLIP [40], PromptAD [41], AdaCLIP [42], and CLIP-AD [43] enhance anomaly representations through prompt learning and cross-modal adaptation. FreqPrompt-AD [5], AnomalyLVM [6], DSSM-MAPL [7], and RTVP [44] further improve local anomaly modeling and vision-language interaction. However, these methods remain limited to anomaly detection and have not extended to anomaly understanding.

2.3 Multi-modal Anomaly Understanding

As research moves from anomaly detection to anomaly understanding, AnomalyGPT [13] encodes local anomaly information into prompt embeddings for anomaly description and question answering. Anomaly-OV [23] strengthens anomaly visual tokens for fine-grained understanding. These methods mainly construct internal anomaly knowledge from the query image, but lack explicit modeling of the structural relation between normal and anomalous states. Some methods further introduce external knowledge. AnomalyR1 [12] combines the query image with domain text knowledge. JUDO [11] incorporates normal images and domain knowledge for anomaly reasoning. However, external normal references are often directly introduced without explicitly modeling their relevance to the query image. Overall, existing methods remain limited in both internal and external knowledge utilization: internal knowledge lacks explicit normal-to-anomaly relation modeling, while external knowledge cannot ensure structurally matched normal references. Unlike existing methods, AD-Seeker++ focuses on the joint modeling of internal and external knowledge. Internal knowledge explicitly captures normal patterns and their structural relations to anomalous states. External knowledge models the relevance between the query image and normal references to obtain better-matched visual cues. The two complementary knowledge sources jointly support fine-grained industrial anomaly cognition.

3 Multi-Modal Industrial Knowledge Benchmark

3.1 Motivation

Although Multi-Modal Large Language Models (MLLMs) have enhanced visual understanding in Industrial Anomaly Detection (IAD), existing benchmarks still struggle to support systematic anomaly cognition. First, many benchmarks [14], [23] are constructed by integrating existing datasets or collecting data in controlled environments. Despite increasing data scale, the diversity of object categories and visual distributions remains limited. Second, existing benchmarks typically evaluate Multiple-Choice Questions (MCQ) or open anomaly descriptions as independent tasks (Tab. 2 and Fig. 1). They lack a continuous process spanning Scene Knowledge Pretraining, Anomaly Understanding, and OpenQA Cognition. Here, OpenQA cognition refers to the ability to understand normal patterns, identify and explain deviations from these patterns, and make inspection judgments without predefined candidate answers. To address these challenges, we construct the Multi-Modal Industrial Knowledge Benchmark-400K (MMIK-400K). It contains 461,388 real industrial images across 432 object categories, with 1,271,155 closed QA pairs and 133,140 open QA pairs. MMIK-400K introduces new objects and environmental variations from real-world industrial scenarios. This expands the anomaly distributions beyond those covered by existing benchmarks. MMIK-400k also establishes a continuous process from normal-state understanding and anomaly deviation analysis to inspection judgments. To the best of our knowledge, MMIK-400K is the first and largest-scale IAD benchmark to unify Scene Knowledge Pretraining, Anomaly Understanding, and OpenQA Cognition. It provides a comprehensive foundation for knowledge learning and cognition evaluation in multi-modal IAD.

3.2 Data Collection and Annotation

3.2.1 *Multi-Source Data Aggregation and Standardization*

As shown in Fig. 2-A, we first collect 43 new foundational industrial datasets from real-world environments. This aims to enable the model to learn a general industrial anomaly feature space. These datasets denoted as $\mathcal{D}_{raw} = \{x_i\}_{i=1}^{N}$, where the total number of samples is $N \approx 5\times10^5$. However, real-world images often contain complex environmental noise. To reduce interference from peripheral backgrounds, we introduce spatial localization cropping $\mathcal{F}_{crop}$. We define cropping windows based on the image characteristics of each data to extract valid image regions. For any query image x_i, the extraction of its core feature region is formulated as:

$$x_{roi}^{(i)} = \mathcal{F}_{crop}(x_i, \mathbf{b}_i), \tag{1}$$

where $\mathbf{b}_i$ denotes the geometric bounding box parameters determined by experts. After extracting the core anomaly

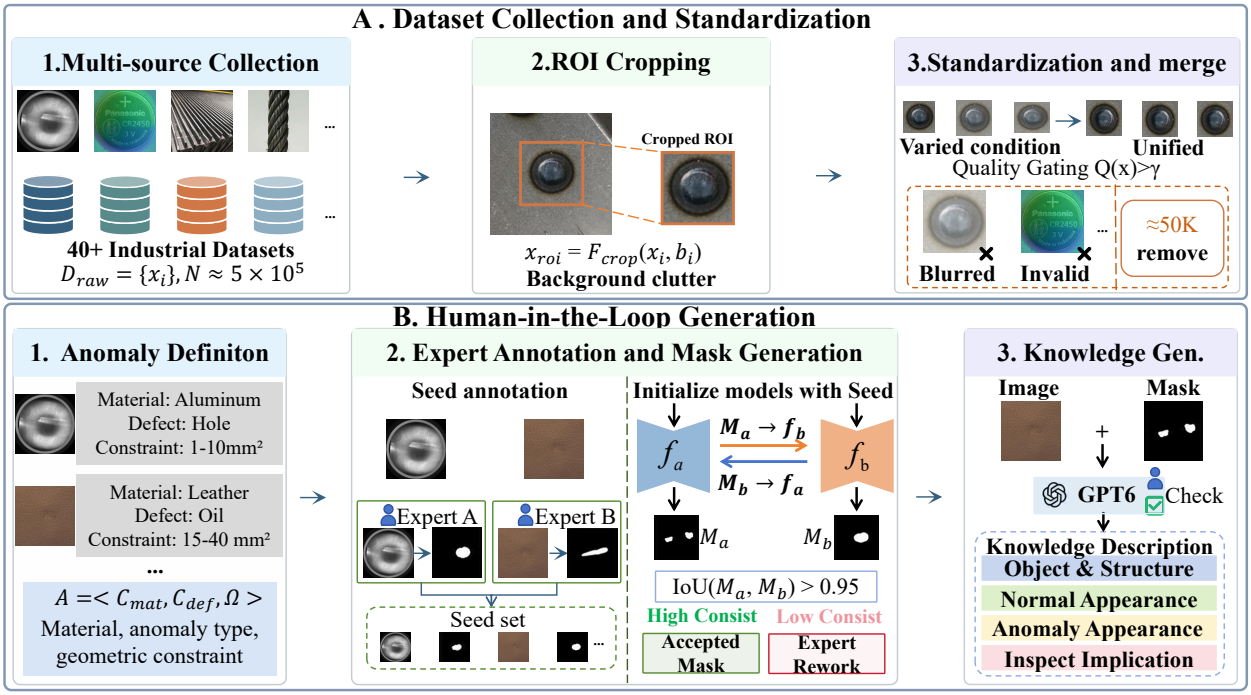

Fig. 2. Data Collection and Annotation Pipeline. (a) We collect 43 foundational industrial datasets and perform multi-source data aggregation and standardization. (b) Human collaborative mask generation. We design a dual-branch model for semi-automated annotation to improve mask quality.

regions, we observe substantial differences in image resolution and illumination shifts in some images. Therefore, we perform feature alignment and normalization on the cropped images. Specifically, we apply a standardization projection $\mathcal{F}_{std}$, which incorporates spatial-depth resampling and pixel-level normalization. To ensure the purity of the benchmark, we introduce a quality gate $\mathcal{Q}(\cdot)$ and manually remove approximately 45,000 low-quality images. The resulting purified and merged normalized dataset is formally defined as:

$$\mathcal{D}_{merged} = \left\{ \mathcal{F}_{std}(x_{roi}^{(i)}) \mid \mathcal{Q}(x_{roi}^{(i)}) > \gamma \right\}, \tag{2}$$

where $\mathcal{Q}(\cdot)$ denotes the image quality scoring function, and γ is the quality screening threshold. Finally, we construct a self-collected raw anomaly dataset containing 461,388 images through multi-source aggregation, standardization, and quality screening.

3.2.2 *Human-in-the-Loop Annotation and Knowledge Construction*

For large-scale dense pixel-level annotation, manual labeling is not only inefficient but also prone to introduce label noise. Therefore, we propose a Human-in-the-Loop method. First, we transform anomaly determination from subjective visual experience into a set of constrained parameters. As shown in Fig.2-b, each anomaly instance is formalized as a triplet:

$$\mathcal{A} = \langle \mathcal{C}_{mat}, \mathcal{C}_{def}, \Omega \rangle. \tag{3}$$

where $\mathcal{C}_{mat}$ denotes the material category, $\mathcal{C}_{def}$ represents the category, and Ω defines the anomaly constraints. Under these constraints, domain experts perform high-fidelity initial pre-annotation. After obtaining the seed dataset, we design a cross-validation mechanism to improve the efficiency of large-scale mask generation. Specifically, we first pre-train two independent mask generation networks, f_a and f_b, using the seed dataset. When processing large-scale unlabeled data, we use the pseudo-labels generated by f_a to fine-tune f_b, and vice versa. This cross-validation training mechanism helps overcome potential prediction errors arising from single-model training. During the final mask generation and validation stage, the two models independently predict masks M_a and M_b for query image x. We use Intersection over Union (IoU) to quantify the consistency between their anomaly predictions:

$$\text{IoU}(M_a, M_b) = \frac{|M_a \cap M_b|}{|M_a \cup M_b|}, \tag{4}$$

where M_a and M_b denote the predicted masks of the two networks, respectively. Based on this metric, we apply the following gating rule to validate the predicted masks:

$$\mathcal{Y}_{mask} = \begin{cases} \Phi(M_a, M_b), & \text{if IoU}(M_a, M_b) > 0.95 \quad \text{(Yes)}, \\ \mathcal{H}_{expert}(x \mid \Omega), & \text{otherwise} \quad \text{(Rework)}, \end{cases} \tag{5}$$

where, $\mathcal{Y}_{mask}$ denotes the mask output, Φ represents the mask fusion operation, and $\mathcal{H}_{expert}$ denotes expert re-annotation under the constraints Ω. When the two predictions are highly consistent, the resulting mask passes cross-validation and undergoes final expert inspection. Otherwise, the uncertain sample is returned for expert re-annotation. To assess annotation quality, two industrial inspection experts, each with one year of experience, re-annotate automatically labeled images sampled from 432 categories. As shown in Table 1, the automatic annotations achieve an IoU of at least 0.95 against expert annotations. This result supports the reliability of automatic annotations.

TABLE 1
Quality assessment of automatic annotations.

Metric	Value	Notes
IoU ≥ 0.95	0.954	Mask IoU between auto. and expert.
Precision@IoU ≥ 0.5	0.959	Automatic category accuracy.
Recall@IoU ≥ 0.5	0.943	Automatic category recall.

After obtaining reliable pixel-level annotations, we further construct image-level knowledge descriptions. As shown in Fig. 2-B, each image and its corresponding mask are fed into GPT-5 [45]. A unified prompt template is used to generate structured knowledge covering four aspects: (1) object categories, structural composition, and component relationships. (2) normal surface textures, colors, shapes, and integrity. (3) anomaly categories, locations, shapes, textures, and local structural changes, and (4) quality impacts, structural risks, and inspection implications. Finally, experts verify and correct ambiguous, or visually unsupported descriptions. This process produces image-knowledge pairs with consistent visual and textual information.

3.3 Structured Knowledge Architecture

To address the lack of evaluation from understanding to cognition in existing IAD benchmarks, we introduce a structured knowledge-driven benchmark. MMIK-400K enables models to first learn general industrial knowledge, including object structures, surface states, and spatial relationships. It then supports the understanding of normal states, anomalous deviations, and inspection judgments. Retrievable normal visual knowledge further provides dynamic references for inference. Retrievable normal visual knowledge further provides dynamic references for inference. Unlike simple dataset aggregation, MMIK-400K emphasizes the progression from anomaly understanding to cognition. MMIK-400k establishes a complete pipeline from knowledge pretraining to anomaly understanding-cognition evaluation. The overall benchmark is formulated as:

$$\mathcal{B}_{\mathrm{MMIK}} = \{\mathcal{D}_{\mathrm{SKU}}, \mathcal{D}_{\mathrm{IAC}}, \mathcal{I}_{\mathrm{RAG}}\}, \tag{6}$$

where $\mathcal{D}_{\mathrm{SKU}}$ denotes the Scene Knowledge Understanding dataset, $\mathcal{D}_{\mathrm{IAC}}$ denotes the Industrial Anomaly Cognition dataset, and $\mathcal{I}_{\mathrm{RAG}}$ represents the RAG-based visual knowledge. Thus, MMIK-400K advances industrial anomaly evaluation from simple question-answer to open cognitive evaluation.

3.3.1 Scene Knowledge Understanding

MMIK-400K first establishes a large-scale Industrial Scene Knowledge Understanding (SKU) pretraining subset. SKU contains 417,008 images, all of which are anomalous samples, across 337 industrial categories. SKU provides 960,495 closed question-answer pairs and is used exclusively for large-scale industrial knowledge pretraining. All questions follow a multiple-choice format with five predefined options. They provide unified scene knowledge supervision covering anomaly presence, category, location, quantity, and overall judgment. Unlike benchmarks such as MMAD [14] and MANTA [15], which primarily use multiple-choice questions for anomaly understanding, SKU employs them as a vehicle for large-scale knowledge pretraining. This enables models to acquire foundational industrial scene knowledge before anomaly cognition. Based on image categories, anomaly annotations information, we further construct five scene knowledge tasks:

$$\mathcal{T}_{\mathrm{SKU}} = \{\mathrm{AD}, \mathrm{ACls}, \mathrm{AL}, \mathrm{ACnt}, \mathrm{AA}\}, \tag{7}$$

where AD, ACls, AL, ACnt, and AA denote Anomaly Detection, Anomaly Classification, Anomaly Localization, Anomaly Counting, and Anomaly Analysis, respectively. As shown in Fig. 3, the first four tasks focus on anomaly presence, category, location, and quantity, respectively. Anomaly Analysis integrates this information to form an overall inspection judgment. Thus, SKU transforms scattered visual annotations into semantically connected scene knowledge supervision for industrial knowledge pretraining, rather than evaluating individual task performance, thereby providing a foundation for subsequent anomaly cognition.

3.3.2 Industrial Anomaly Cognition

SKU provides models with foundational industrial Understanding. However, further anomaly cognition requires building normal pattern, anomalous manifestations, and inspection judgments. Thus, MMIK-400K introduces the Industrial Anomaly Cognition (IAC) subset for fine-tuning and formal evaluation after SKU pretraining. IAC QA is constructed independently of SKU, with no sample overlap.

IAC contains 44,380 industrial object images across 95 anomaly categories, including 38,410 normal images and 5,970 anomalous images. For anomalous samples, we further provide masks, categories, spatial locations, and indices of normal visual knowledge from the same category. These indices provide the foundation for subsequent knowledge retrieval. IAC comprises both closed Knowledge Understanding (ClosedQA) and open Anomaly Cognition (OpenQA). ClosedQA uses predefined options to establish structured relationships among object attributes, anomaly categories, spatial locations, visual appearances, and inspection implications. In contrast, OpenQA removes candidate constraints to evaluate autonomous anomaly cognition. Unlike existing benchmarks that separately emphasize multiple-choice questions, knowledge descriptions, or chain-of-thought (CoT) reasoning, IAC integrates ClosedQA and OpenQA into a continuous process. Models first acquire foundational industrial knowledge and then understand anomaly categories, locations, appearances, and impacts. Finally, they identify deviations from normal states and form inspection judgments. ClosedQA comprises seven tasks divided into anomaly understanding and object understanding:

$$\begin{aligned}
&\mathcal{T}_{\mathrm{Anomaly}} = \{\mathrm{Detection}, \mathrm{Classification}, \mathrm{Localization}, \mathrm{Description}, \mathrm{Analysis}\},\\
&\mathcal{T}_{\mathrm{Object}} = \{\mathrm{Classification}, \mathrm{Analysis}\},\\
&\mathcal{T}_{\mathrm{ClosedQA}} = \mathcal{T}_{\mathrm{Anomaly}} \cup \mathcal{T}_{\mathrm{Object}}.
\end{aligned} \tag{8}$$

where, $\mathcal{T}_{\mathrm{Anomaly}}$ and $\mathcal{T}_{\mathrm{Object}}$ denote the anomaly understanding and object understanding task sets, respectively. Their union forms the complete ClosedQA task set $\mathcal{T}_{\mathrm{ClosedQA}}$.

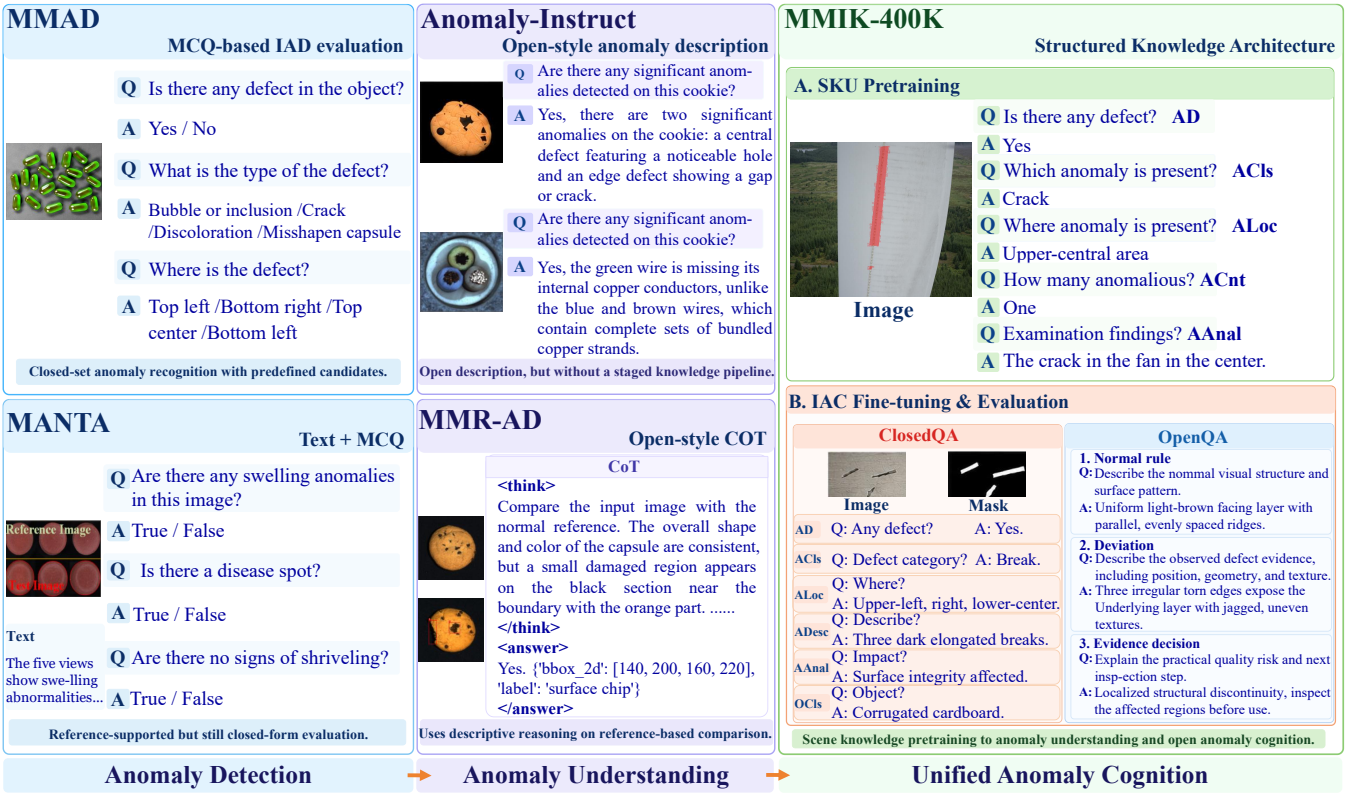

Fig. 3. Comparison of task paradigms between mainstream multi-modal IAD benchmarks and MMIK-400K. MMIK-400K first unifies Scene Knowledge Pretraining, Anomaly Understanding, and OpenQA Cognition into a continuous process.

We refer to existing multiple-choice IAD baselines [14], [15]. Anomaly understanding covers anomaly presence, category, location, visual appearance, and potential impacts. Object understanding focuses on object categories and overall state judgments. Each image is associated with seven ClosedQA tasks, yielding 310,660 closed question-answer pairs. However, ClosedQA remains constrained by predefined options, allowing models to rely on candidate semantics. To address this limitation, we introduce OpenQA to evaluate autonomous anomaly cognition. Without predefined options, models must understand normal patterns, explain anomalous deviations, and form inspection conclusions. We construct three types of OpenQA for each image:

$$\mathcal{T}_{\text{OpenQA}} = \{\text{normal_rule}, \text{deviation}, \text{evidence_decision}\}. \tag{9}$$

The **Normal_rule** task requires models to summarize the structural and appearance patterns of objects under normal conditions. The **Deviation** task requires models to explain deviations in location, morphology, or structural integrity based on the current image and anomaly region. The **Evidence_decision** task further requires models to make inspection or maintenance judgments. These tasks form a complete anomaly cognition process, comprising 133,140 open question-answer pairs.

3.3.3 Image Knowledge Index

To support general visual knowledge learning, we construct an image-level knowledge for MMIK-400K. MMIK-400K introduces retrievable normal visual knowledge during inference. MMIK-400K incorporates retrievable normal visual knowledge during the inference. For each category of anomalous images, we provide a corresponding set of 8 normal images to serve as a retrieval pool.

3.3.4 Evaluation Metrics

MMIK-400K evaluates anomaly understanding and open anomaly cognition on IAC through ClosedQA and OpenQA, respectively.

ClosedQA Evaluation. ClosedQA comprises seven tasks. We evaluate each task t using accuracy:

$$\text{Acc}_t = \frac{1}{N_t} \sum_{i=1}^{N_t} \mathbb{I}\left(\hat{a}_i^t = a_i^t\right) \times 100, \tag{10}$$

where N_t denotes the number of test samples for task t. $\hat{a}_i^t$ and a_i^t represent the predicted and ground-truth answers for the i-th sample, respectively. The indicator function $\mathbb{I}(\cdot)$ returns 1 for a correct prediction and 0 otherwise. To evaluate overall closed knowledge understanding, we further compute the mean accuracy across the seven tasks:

$$\text{ClosedScore} = \frac{1}{|\mathcal{T}_{\text{Closed}}|} \sum_{t \in \mathcal{T}_{\text{Closed}}} \text{Acc}_t, \tag{11}$$

where $\mathcal{T}_{\text{Closed}}$ denotes the set of all ClosedQA tasks, with $|\mathcal{T}_{\text{Closed}}| = 7$.

OpenQA Evaluation. OpenQA evaluates industrial anomaly cognition through normal-pattern summarization, anomalous-deviation explanation, and inspection judgment, without predefined answer options. Since valid responses may express the same task-relevant content using different

TABLE 2
Comparison of representative industrial anomaly detection and multi-modal understanding benchmarks. MCQ denotes Multi-Choice Question.

Benchmark	Images	Classes	Data Construction	Large Knowledge Pretraining	Anomaly Understanding	OpenQA Cognition Evaluation
MVTec AD [1]	5.4K	15	Controlled	–	–	–
VisA [2]	10.8K	12	Controlled	–	–	–
Real-IAD [3]	150K	30	Controlled	–	–	–
MMAD [14]	8.4K	38	Existing	–	39.7K MCQ	–
MANTA [15]	137K+	38	Controlled	–	2K MCQ + Text	–
Anomaly-Instruct [23]	72K	380	Existing + Web	–	–	125K Instruct (Detection + Description)
MMR-AD [24]	127K	188	Existing	–	–	127K CoT (Detection + Localization)
MMIK-400K	**461K**	**432**	**New + Real-world**	**960K ClosedQA**	**310K ClosedQA**	**133K OpenQA** (Normal Rule + Deviation + Decision)

wording, we evaluate both lexical overlap and semantic agreement. Given a model response $\hat{y}_i$ and reference answer y_i, we employ three complementary metrics: ROUGE-L F1 measures lexical overlap based on the longest common subsequence, BERTScore F1 assesses contextual token-level semantic alignment, and SBERT cosine similarity measures sentence-level semantic agreement. Applied separately to each task, these metrics quantify the agreement between generated responses and reference answers at different linguistic levels, providing a reference-based assessment of performance on the defined anomaly cognition tasks.

3.4 Statistical Analysis of MMIK-400K

Fig. 4 presents representative samples from MMIK-400K, covering diverse industrial scenarios such as metals, electronic components. These categories exhibit substantial differences in appearance, structure, texture, and anomaly patterns, demonstrating the MMIK-400k's scene diversity and distribution coverage.

With 461,388 real-world industrial images across 432 categories, MMIK-400K exhibits pronounced semantic long-tail distributions (Fig. 5), visual heterogeneity, and spatial-morphological uncertainty. These characteristics create complex anomaly distributions beyond existing-object or controlled scenarios. Specifically, Fig. 6-a shows substantial scale variations within individual categories. Fig. 6-b indicates that anomalies of different scales span a wide range of brightness levels, reflecting complex variations in real-world industrial imaging conditions. As shown in Fig. 6-c, 48.5% of images contain multi anomaly types, increasing the difficulty of localization and discrimination in multi-anomaly scenarios. Finally, Fig. 6-d reveals feature overlap across anomaly categories and dispersed distributions within the same category, further challenging fine-grained anomaly detection.

As shown in Tab. 2, MMIK-400K provides broader industrial distribution coverage and a continuous knowledge-to-cognition benchmark. First, several recent benchmarks are constructed from existing public datasets or controlled environments. For example, MMAD [14] integrates four public datasets, while MMR-AD [24] aggregates 14 public datasets. Their object categories and anomaly patterns are constrained by existing data distributions. In contrast, MMIK-400K incorporates new objects, anomaly patterns, and environmental variations from real-world industrial scenarios, covering 461K images across 432 categories. Second, MMIK-400K organizes 1.27M ClosedQA pairs and 133K OpenQA pairs into a continuous process

Fig. 4. Representative samples of MMIK-400K. The samples exhibit substantial diversity in object categories, material properties, structural morphology, and anomaly patterns, demonstrating the broad coverage of diverse real-world industrial distributions.

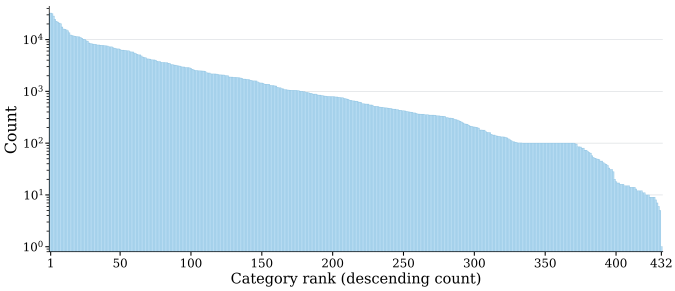

Fig. 5. Distribution of the number of anomalies per-category in MMIK-400K.

from scene knowledge pretraining to anomaly understanding and open anomaly cognition. Specifically, SKU provides 960K ClosedQA pairs for scene knowledge pretraining, while IAC contains 310K ClosedQA pairs for anomaly understanding and 133K OpenQA pairs for open cognition. Unlike existing open-form benchmarks that mainly focus on anomaly description, detection, or localization, OpenQA further evaluates normal-pattern understanding, anomalous deviation explanation, and inspection judgment. Together with visual knowledge retrieval during inference, MMIK-400K establishes a unified framework for industrial anomaly understanding and cognition evaluation.

4 ADSeeker++

4.1 Motivation

Industrial anomaly cognition requires not only anomaly recognition but also reliable explanatory knowledge. These

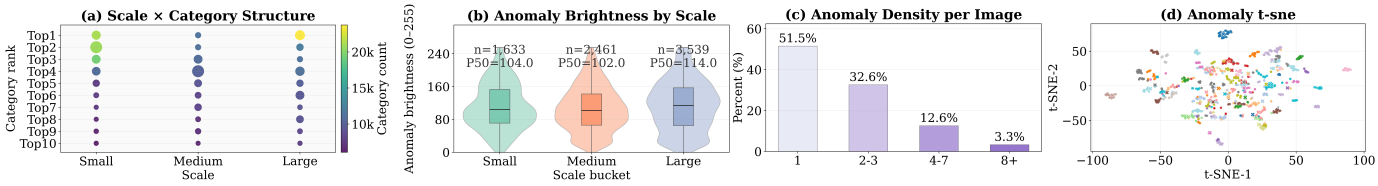

Fig. 6. MMIK-400K presents challenges of category imbalance, scale and brightness variation, uneven anomaly density, and inter-category visual ambiguity.

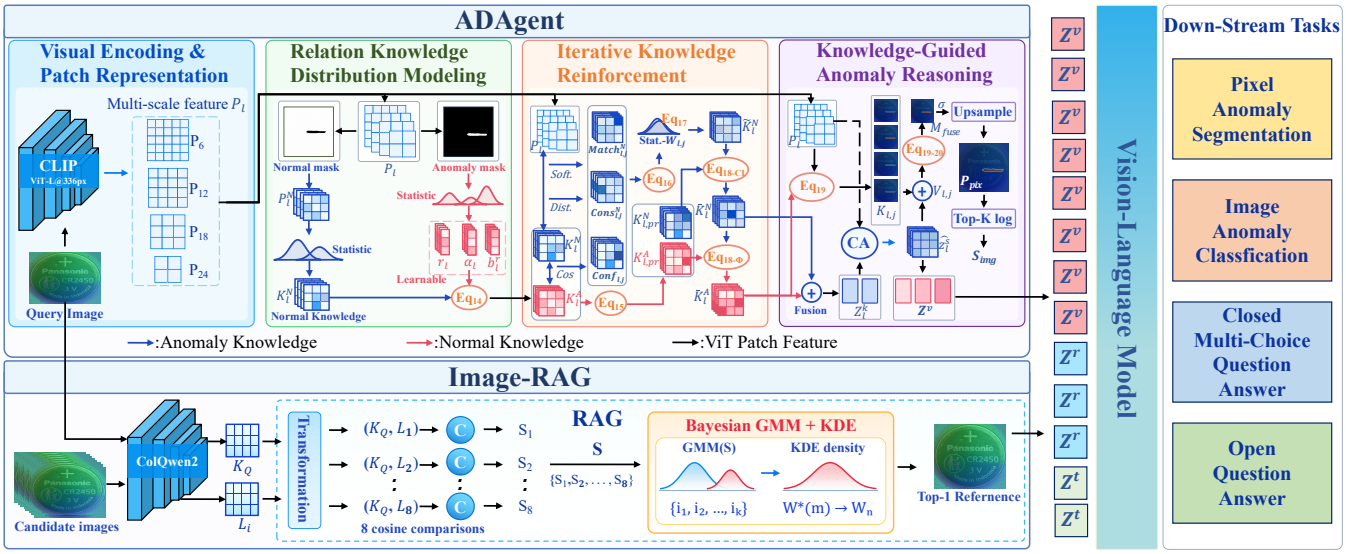

Fig. 7. Overview of ADSeeker++. ADSeeker++ leverages internal knowledge to model normal-to-anomaly relations and incorporates external knowledge retrieval to obtain query image-relevant normal visual references, enabling multi-modal anomaly understanding and reasoning.

knowledge comes from two complementary sources: internal anomaly within the query image and external domain references. However, internal anomalies are often local and subtle, while external retrieval can be dominated by superficial similarity. To address these challenges, we propose **ADSeeker++**. As shown in Fig. 7, ADSeeker++ consists of two complementary advantages: **ADAgent** and **Image-RAG**. ADAgent and I-RAG provide complementary internal and external knowledge, respectively, while the VLM performs collaborative reasoning over both sources for anomaly cognition.

4.2 ADAgent

The key to internal anomaly knowledge learning is to capture not only what normal patterns look like, but also how they deviate toward anomalies. ADAgent adopts a pretrained CLIP ViT-L/14@336px [46] as the visual encoder and extracts patch features from multiple ViT depths.

ADAgent contains three components. As shown in Fig. 7-ADAgent, **Relation Knowledge Distribution Modeling** learns normal knowledge from training images and constructs anomaly knowledge through a learnable normal-to-anomaly relation. **Iterative Knowledge Reinforcement** transfers knowledge by calibrating the normal knowledge while reinforcing the learned relation across ViT depths. **Knowledge-Guided Anomaly Reasoning** finally uses the calibrated normal and anomaly knowledge to guide local anomaly reasoning and produces anomaly feature tokens together with an image-level anomaly score for the subsequent VLM.

4.2.1 *Relation Knowledge Distribution Modeling*

Different ViT depths capture normal appearance and anomaly deviation at different representation levels. Relation Knowledge Distribution Modeling constructs a normal-anomaly knowledge pair at each selected depth $l \in \mathcal{L}$, where $\mathcal{L} = \{6, 12, 18, 24\}$. For the l-th depth, normal-region patch features P_l^N are selected from annotated training images using the projected normal mask. Their weighted feature center and feature-wise variance define the normal knowledge:

$$\mathcal{K}_l^N = (\mu_l^N, v_l^N) = \left(\mathbb{E}_w[P_l^N], \mathrm{Var}_w[P_l^N]\right), \tag{12}$$

where $\mathbb{E}_w[\cdot]$ and $\mathrm{Var}_w[\cdot]$ denote the mean and variance, respectively. μ_l^N describes the center of the normal feature distribution, while v_l^N describes its feature-wise variation. Both are initialized from the training statistics and further optimized during training.

Rather than learning anomaly patterns independently, ADAgent parameterizes how normal knowledge deviates toward anomalies:

$$\begin{aligned} \delta_l^R &= \alpha_l \,\mathrm{Norm}\left(\Pi_{\perp \mu_l^N}(r_l)\right) \odot \sqrt{v_l^N + \epsilon}, \\ v_l^R &= \mathrm{softplus}(b_l^R) + v_{\min}, \\ \mathcal{K}_l^A &= (\mu_l^A, v_l^A) = \left(\mathrm{Norm}\left(\mu_l^N + \delta_l^R\right), v_l^N + v_l^R\right). \end{aligned} \tag{13}$$

Where, r_l and α_l respectively determine the direction and magnitude of the normal-to-anomaly shift, while v_l^R represents its feature-wise uncertainty. $\Pi_{\perp\mu_l^N}(\cdot)$ removes the component aligned with the normal center, $\mathrm{Norm}(\cdot)$ denotes L_2 normalization, and ϵ is a small constant for numerical stability. The $v_{\min}$ is a minimum variance that prevents the relation variance from collapsing to zero. Importantly, r_l, b_l^R, and α_l are initialized using anomaly-region statistics from the training images and are then jointly optimized with the model. In this way, $\mathcal{K}_l^A$ is not an independently memorized defect distribution, but a structured deviation from $\mathcal{K}_l^N$.

4.2.2 Iterative Knowledge Reinforcement

Since different depths provide complementary visual information, Iterative Knowledge Reinforcement progressively integrates these knowledge states across depths. At the l-th selected depth, $\mathcal{K}_l^N$ and $\mathcal{K}_l^A$ denote the normal and anomaly knowledge learned at the current depth. Let l^- denote the preceding selected depth. Then, $\bar{\mathcal{K}}_{l^-}^N$ and $\bar{\mathcal{K}}_{l^-}^A$ denote the knowledge states retained after updating that depth. ADAgent first fuses the current depth-specific knowledge with the accumulated knowledge through Covariance Intersection (CI) [47]:

$$\mathcal{K}_{l,\mathrm{pr}}^s = \mathrm{CI}\left(\mathcal{K}_l^s, \bar{\mathcal{K}}_{l^-}^s\right), \qquad s \in \{N, A\}. \tag{14}$$

Where, $\mathcal{K}_{l,\mathrm{pr}}^s$ denotes the knowledge state after cross-depth fusion at depth l. The subscript pr indicates that it serves as the prior for the current-depth update. At the first selected depth, no preceding accumulated state is available, and thus $\mathcal{K}_{l,\mathrm{pr}}^s = \mathcal{K}_l^s$. CI uses precision-weighted means and diagonal variances, with weights determined by learning. Normal centers are normalized after fusion, while relation offsets retain their magnitude. The recursion is reset for each query image.

The patch features $P_l = \{p_{l,j}\}$ are then used to further reinforce the normal knowledge. The weight of each patch is jointly determined by its matching with the normal knowledge, the discrimination confidence between normal and anomaly knowledge, and its consistency with the normal center:

$$w_{l,j} = \left(\mathrm{Match}_{l,j}^N\right)^2 \left(0.25 + 0.75\,\mathrm{Conf}_{l,j}\right) \mathrm{Cons}_{l,j}^N. \tag{15}$$

Where, $\mathrm{Match}_{l,j}^N$ is obtained by applying Softmax to the variance normalized distances from the patch to the normal knowledge. $\mathrm{Conf}_{l,j}$ is determined based on the degree of separation between normal and anomalous distances, while $\mathrm{Cons}_{l,j}^N$ measures the cosine consistency between the patch and the normal center. Therefore, patches that better match the normal knowledge receive larger weights. The weighted patch statistics form the normal knowledge:

$$\widetilde{\mathcal{K}}_l^N = \left(\mathbb{E}_w[P_l], \mathrm{Var}_w[P_l]\right), \tag{16}$$

where $\mathbb{E}_w[\cdot]$ and $\mathrm{Var}_w[\cdot]$ denote the weighted mean and variance according to $w_{l,j}$, respectively. This representation retains reliable normal characteristics of the query image while suppressing the influence of anomalous or uncertain patches. Finally, a second CI operation incorporates the current normal representation into the cross-depth normal knowledge. The anomaly knowledge is reconstructed from the updated normal knowledge:

$$\begin{cases} \bar{\mathcal{K}}_l^N = \mathrm{CI}\left(\mathcal{K}_{l,\mathrm{pr}}^N, \widetilde{\mathcal{K}}_l^N\right), \\ \bar{\mathcal{K}}_l^A = \Phi\left(\bar{\mathcal{K}}_l^N, \mathcal{K}_{l,\mathrm{pr}}^A\right). \end{cases} \tag{17}$$

Where, $\bar{\mathcal{K}}_l^N$ and $\bar{\mathcal{K}}_l^A$ denote the reinforced normal and anomaly knowledge after processing depth l, respectively. $\Phi(\cdot)$ denotes the transformation defined in Eq. (13), which reconstructs the anomaly state from the updated normal state and the accumulated normal-to-anomaly relation. The resulting $\bar{\mathcal{K}}_l^N$ and $\bar{\mathcal{K}}_l^A$ are then retained as the accumulated knowledge states for the next selected depth.

4.2.3 Knowledge-Guided Anomaly Reasoning

ADAgent combines knowledge matching and token-guided visual scoring for patch-level anomaly prediction. At depth l, normalized patch features are matched with the reinforced normal and anomaly knowledge using variance-normalized distances. A softmax over the negative, temperature-scaled distances yields the normal and anomaly probabilities $\bar{q}_{l,j}^N$ and $\bar{q}_{l,j}^A$. Their log-odds define the knowledge-guided anomaly response:

$$K_{l,j} = \log \frac{\bar{q}_{l,j}^A + \epsilon}{\bar{q}_{l,j}^N + \epsilon}, \tag{18}$$

where ϵ is a small constant for numerical stability. The two distances share the pooled normal/anomaly diagonal variance. A larger $K_{l,j}$ indicates stronger anomaly evidence. The reinforced normal and anomaly knowledge also enhance their corresponding visual tokens through feature fusion $\mathcal{F}_{\mathrm{RU}}^s$. The enhanced tokens aggregate spatial information from the current-depth patches through Cross-Attention (CA). Their relative cosine responses then provide visual anomaly score for each patch:

$$\begin{aligned} \mathbf{Z}_l^{K,s} &= \mathcal{F}_{\mathrm{RU}}^s\left(\mathbf{z}^s, \bar{\mathcal{K}}_l^s\right), \\ \widehat{\mathbf{z}}_l^s &= \mathrm{CA}_l\left(\mathbf{Z}_l^{K,s}, P_l\right), \quad s \in \{N, A\}, \\ V_{l,j} &= \cos\left(\widetilde{p}_{l,j}, \widehat{\mathbf{z}}_l^A\right) - \cos\left(\widetilde{p}_{l,j}, \widehat{\mathbf{z}}_l^N\right). \end{aligned} \tag{19}$$

Where, $\mathbf{z}^s$ is the original visual token, $\mathbf{Z}_l^{K,s}$ is its knowledge-enhanced representation, and $\widehat{\mathbf{z}}_l^s$ is the adapted token returned by CA. The feature $\widetilde{p}_{l,j}$ is obtained by applying the shape transformation at depth l to patch j. The visual response $V_{l,j}$ measures the patch's relative similarity to the anomaly and normal tokens.

The visual response $V_{l,j}$ and knowledge response $K_{l,j}$ are combined. Local affinity refinement of the fused response produces the patch anomaly probabilities. These probabilities are arranged on the patch grid to form the layer-wise anomaly probability map $\widehat{P}_l$. Predictions from different depths are subsequently fused in logit space:

$$\begin{aligned} M_{\mathrm{fuse}} &= |\mathcal{L}| \sum_{l \in \mathcal{L}} \pi_l \odot \mathrm{logit}(\widehat{P}_l), \\ P_{\mathrm{fuse}} &= \sigma(M_{\mathrm{fuse}}), \end{aligned} \tag{20}$$

where π_l is a nonnegative spatial weight normalized across the selected depths. The weights reflect local response quality and depth priors. The factor $|\mathcal{L}|$ preserves the layer-

TABLE 3
Comparison with zero-shot anomaly detection methods on **MVTec AD, VisA, BTAD, and DTD**. Each entry reports I-AUROC / I-AP | P-AUROC / PRO (%). Bold and underlined values indicate the best and second-best results, respectively.

Dataset	ADAgent	PILOT [48]	VisualAD-DINO [49]	CLIP-AD$^+$ [43]	AnomalyCLIP [40]	AdaCLIP [42]	FAPrompt [50]
MVTec	**92.9** / **97.4** | **91.3** / 88.4	90.2 / 95.7 | 89.4 / 79.2	90.1 / 94.8 | **91.3** / **88.6**	92.2 / 96.6 | 91.2 / 85.1	91.5 / 96.2 | 91.1 / 81.4	90.0 / 95.7 | 89.9 / 44.1	91.9 / 95.7 | 90.6 / 83.3
VisA	85.0 / **87.9** | **96.4** / **91.4**	82.0 / 84.1 | 94.8 / 85.6	83.1 / 86.8 | 95.3 / 88.2	78.3 / 82.0 | 94.0 / 83.0	82.1 / 85.4 | 95.5 / 87.0	**85.8** / 87.6 | 95.8 / 51.3	84.6 / 86.8 | 95.9 / 87.7
BTAD	**95.1** / **97.5** | 93.3 / **81.8**	90.0 / 91.6 | 94.9 / 73.0	88.2 / 89.7 | 93.4 / 76.7	82.4 / 81.3 | 93.3 / 66.9*	88.3 / 87.3 | 94.2 / 74.8	90.0 / 93.8 | 93.7 / 20.2	92.2 / 92.5 | **95.6** / 75.1
DTD	**98.7** / **99.8** | **99.1** / **96.3**	95.3 / 98.0 | 97.3 / 89.9	91.0 / 97.4 | 96.7 / 92.4	95.7 / 98.5 | 98.6 / 94.5	93.5 / 97.0 | 97.9 / 92.3	91.3 / 95.4 | 96.9 / 68.1	96.2 / 98.1 | 98.3 / 93.3

sum scale, σ denotes sigmoid, and $\odot$ denotes element-wise multiplication. The fused probability map P_{fuse} is used forpixel-level supervision.

At inference, the fused logits M_{fuse} are upsampled to the query image resolution and Gaussian-smoothed to obtain the pixel-level anomaly score map. The image-level anomaly score is computed by averaging its highest-response entries:

$$\begin{aligned} P_{\text{pix}} &= \text{Smooth}\left(\text{Upsample}(M_{\text{fuse}})\right), \\ S_{\text{img}} &= \text{TopKMean}(P_{\text{pix}}). \end{aligned} \tag{21}$$

Where, P_{pix} denotes the pixel-level anomaly response map, and S_{img} represents the anomaly score.

4.2.4 Optimization Objective

To jointly optimize image-level anomaly discrimination and pixel-level localization, ADAgent employs multi-granularity anomaly supervision. At the image level, high-response regions of the fused anomaly map are aggregated into the image-level anomaly score S_{img}, which is supervised by binary cross-entropy:

$$\mathcal{L}_{\text{img}} = \text{BCEWithLogits}\left(S_{\text{img}}, y_{\text{img}}\right), \tag{22}$$

where S_{img} denotes the Top-K averaged logit of the fused anomaly map, and $y_{\text{img}} \in \{0, 1\}$ is the image-level anomaly label.

At the pixel level, both the fused anomaly map and the depth-wise predictions are explicitly supervised. We combine Focal and Dice losses to account for pixel-wise discrimination and anomaly-region overlap:

$$\begin{aligned} \mathcal{L}_{\text{seg}} &= 0.70\,\ell_{\text{map}}(P_{fuse}, y) + \frac{0.30}{|\mathcal{L}|}\sum_{l\in\mathcal{L}}\ell_{\text{map}}(P_l, y), \\ \ell_{\text{map}}(p, y) &= \ell_{\text{focal}}(p, y) + \ell_{\text{dice}}(p, y), \end{aligned} \tag{23}$$

where P_{fuse} and P_l denote the fused anomaly map and the anomaly map predicted at the l-th ViT depth, respectively, and y is the pixel-level anomaly annotation. We assign 0.70 of the supervision weight to the fused prediction, while the remaining 0.30 is uniformly distributed across the selected ViT depths. This design emphasizes the final anomaly prediction while preserving anomaly sensitivity at different representation depths.

4.3 Image-RAG

Instead of retrieving textual documents, Image-RAG retrieves normal images and provides them as visual context for VLM reasoning. As shown in Fig. 7-Image-RAG, For each query image, we construct a category-specific candidate set containing eight normal images and retrieve the normal reference that is most consistent with the current appearance.

Given a query image and candidate images, they are uniformly encoded by ColQwen2 into K_Q and $L = \{L_i\}$ (where $i = 8$), and their relevance is measured by cosine similarity:

$$\mathcal{S}(K_Q, L) = \{S_i \mid S_i = \cos(K_Q, L_i),\ L_i \in L\}. \tag{24}$$

The resulting similarities characterize how well each normal candidate matches the query image. However, normal candidates from the same category can exhibit highly similar appearances, causing their similarity scores to concentrate within a narrow range. To better distinguish such candidates, Image-RAG further models the distribution of the similarity scores.

We assume that similarity groups approximately follow Gaussian distributions:

$$N = \left\{I_n \sim \mathcal{G}(\mu_n, \sigma_n^2) \mid I_n \in \mathcal{S}\right\}, \tag{25}$$

where $\mathcal{G}$ denotes a Gaussian distribution, and μ_n and σ_n^2 are the corresponding mean and variance. Based on the ranked similarity scores $\mathcal{S}$, Bayesian Gaussian Mixture Modeling [51] is used to estimate the underlying similarity groups:

$$\text{GMM}(\mathcal{S}) = [i_1, i_2, \ldots, i_K \mid i \sim \mathcal{G}(\mu, \sigma^2)]. \tag{26}$$

Kernel Density Estimation [52] is then applied to characterize the density of each similarity group. The corresponding group weight is computed as:

$$W_n = \frac{1}{K}\sum_{m\in i_n}\exp\left(W^*(m) - \max_n W^*(n)\right), \tag{27}$$

where W^* denotes the KDE density score. These density-aware weights help distinguish highly concentrated candidates from more informative normal references while preserving their similarity to the query. The query image and the retrieved normal reference are jointly provided to the VLM, allowing it to reason about fine-grained anomalies.

4.4 Knowledge Reasoning

After internal anomaly knowledge learning by ADAgent and external reference knowledge retrieval by Image-RAG, the two knowledge sources are integrated only at the VLM input. The internal anomaly tokens $\mathbf{Z}^v$ encode the reinforced anomaly knowledge, while the external knowledge tokens $\mathbf{Z}^r$ provide a retrieved normal visual reference. Together with the task instruction tokens $\mathbf{Z}^t$, they form the unified VLM context:

$$\mathbf{Z} = \left[\mathbf{Z}^v; \mathbf{Z}^t; \mathbf{Z}^r\right]. \tag{28}$$

TABLE 4
Anomaly detection results on **MMIK-400K**.

Method	I-AUROC	I-AP	P-AUROC	PRO
ADAgent	**83.7**	**82.4**	**88.2**	**69.9**
FAPrompt [50]	75.0	73.9	64.5	37.8
FE-CLIP [53]	77.8	76.4	79.3	61.0
AF-CLIP [54]	80.6	78.9	81.2	63.5
PILOT [48]	72.0	70.6	72.7	55.6
VisualAD [49]	81.3	80.8	79.4	61.3
AnomalyCLIP [40]	50.7	50.6	57.6	25.5
CLIP-AD$^+$/SDP$^+$ [43]	62.6	61.3	63.4	38.1
AdaCLIP [42]	70.3	68.7	61.8	35.7

TABLE 5
Seven-task accuracy (%) on **MMAD** [14]. Paired entries report defect classification/ localization, defect description/analysis, and object classification/ analysis.

Method	Anomaly Disc.	Defect Cls./Loc.	Defect Desc./Ana.	Object Cls./Ana.	Avg.
Random Chance	50.00	25.00/25.00	25.00/25.00	25.00/25.00	28.57
Human Ordinary	86.90	66.25/85.58	71.25/81.52	89.58/69.72	78.69
Human Expert	95.24	75.00/92.31	83.33/94.20	86.11/80.37	86.65
Claude-3.5-Sonnet [58]	60.14	60.14/48.81	67.13/79.11	85.19/79.83	68.62
Gemini-2.5-Flash [59]	**93.38**	69.88/63.30	76.41/81.57	**94.04**/82.00	80.08
GPT-5-mini [45]	64.10	67.35/69.07	79.02/86.72	93.96/83.37	77.66
Qwen2.5-VL-7B [56]	71.39	54.35/61.17	65.81/79.32	91.44/84.43	72.56
LLaVA-OneVision-7B [60]	51.77	46.13/41.85	62.19/69.73	90.31/80.93	63.27
InternVL3.5-8B [61]	67.50	49.37/57.90	58.07/77.66	72.01/81.11	66.23
Kimi-VL-A3B-16B [62]	72.93	53.49/59.66	72.39/81.74	91.91/85.89	74.00
MiMo-VL [63]	54.45	59.56/60.77	71.50/78.48	90.56/81.60	70.99
AnomalyGPT [13]	65.57	27.49/27.97	36.86/32.11	29.84/35.82	36.52
EIAD [64]	60.50	50.70/55.60	67.80/76.90	91.80/82.70	69.43
ADSeeker-3B [16]	58.17	48.71/53.67	69.79/80.36	86.78/82.25	68.53
AnomalyR1-7B [12]	60.93	64.81/70.72	79.06/85.52	93.12/86.91	77.30
AD-Copilot-7B [65]	73.64	67.89/64.08	80.60/85.91	91.06/87.78	78.71
JUDO-7B [11]	65.04	**74.74**/73.01	84.56/89.41	**94.04**/87.58	**81.20**
ADSeeker++-4B	50.35	69.69/**74.50**	**88.43**/**90.99**	92.61/**91.26**	79.69

5 EXPERIMENTS

5.1 Dataset and Evaluation Metrics

We comprehensively evaluate ADSeeker++ on six datasets from three aspects: **anomaly detection, anomaly understanding, and anomaly cognition**. For anomaly detection, MVTec AD [22], VisA [2], BTAD [19], and DTD-Synthetic [55] are adopted for evaluation. At the image level, we report AUROC, and AP to measure anomaly recognition capability. At the pixel level, we report AUROC, and PRO to evaluate anomaly localization accuracy. For anomaly understanding, we evaluate the visual question answering ability on MMIK-400K ClosedQA and MMAD [14], following the ClosedQA Evaluation metrics defined in Sec. 3.3.4. For anomaly cognition, we use MMIK-400K OpenQA to evaluate open anomaly reasoning ability. The evaluation covers three types of questions, including Deviation, Evidence Decision, and Normal Rule, and follows the OpenQA Evaluation metrics defined in Sec. 3.3.4.

5.2 Implementation Details

The experiments are conducted on Ubuntu 22.04.1 with Python 3.10.19 and CUDA Runtime 12.4. The hardware platform consists of three NVIDIA RTX A6000 GPUs. For ADAGent, we adopt CLIP ViT-L/14@336px as the visual backbone with an query image resolution of 518. The ViT backbone is frozen, and only the modules described in Sec. 4.2.1-4.2.3 are optimized. We train ADAGent using AdamW with a batch size of 8 for 3 epochs and a weight decay of 0.01. The first epoch is used for knowledge warm-up with a learning rate of 3×10^{-4}, while the learning rate is reduced to 1×10^{-4} for the remaining two epochs. For Image-RAG, we construct an image retrieval index using ColQwen2 [56] and retrieve the Top-1 image from the predefined normal reference candidates as external visual knowledge. For VLM fine-tuning, we adopt Qwen3-VL-4B-Instruct [57] and apply LoRA only to the $q/k/v/o$ projections and the MLP gate/up/down projections of the last 30 language-model layers, while keeping the visual encoder frozen. The LoRA configuration uses $r = 32$, $\alpha = 32$, dropout $= 0.05$, and bias=none. The model is trained for 2 epochs with 29,588 steps using BF16 precision. The single-GPU batch size is set to 1, with a learning rate of 6×10^{-6}, warm-up ratio of 0.05, gradient clipping of 0.5, and weight decay of 0.

5.3 Quantitative experiments with the State-of-the-art

5.3.1 *Anomaly Detection quality analysis*

We first evaluate ADAgent for zero-shot anomaly detection. Following the cross-dataset setting, the model is trained on MVTec AD [1] and evaluated on VisA [2], BTAD [19], and DTD-Synthetic [55]. For MVTec AD evaluation, VisA is used as the training set. As shown in Tab. 3, Different from FAPrompt [50], which separates normal and anomaly prompts in the text space, and VisualAD [49], which learns normal and anomaly visual tokens, ADAgent represents normal knowledge as visual feature distributions and explicitly models the structural shift from normal to anomaly. On MVTec AD, ADAgent improves I-AUROC and I-AP by 0.7% and 0.8% over the best competitors, while achieving comparable P-AUROC. On VisA, ADAgent improves P-AUROC and PRO by 0.5% and 3.2%, respectively. On BTAD, ADAgent obtains 2.9% and 5.1% gains in I-AUROC and PRO. Moreover, ADAgent achieves the best performance on all four metrics of DTD-Synthetic, demonstrating strong cross-domain transferability.

We further evaluate the generalization ability of ADAgent on MMIK-400K. The training set contains 30,720 normal images and 1,024 anomaly images. The test set contains 3,845 normal images and 4,946 anomaly images. As shown in Tab. 4, existing methods achieve only 57.6%-81.2% P-AUROC and 25.5%-63.5% PRO, indicating that they struggle to distinguish normal variations from structural deviations under complex industrial appearances in MMIK-400K. In contrast, ADAgent achieves 83.7% I-AUROC, 82.4% I-AP, 88.2% P-AUROC, and 69.9% PRO, improving over the best competitors by 2.4%, 1.6%, 7.0%, and 6.4%, respectively. The significant gains in pixel-level metrics demonstrate that ADAgent better models normal visual distributions and local anomaly deviations, leading to more accurate anomaly localization in complex industrial scenarios.

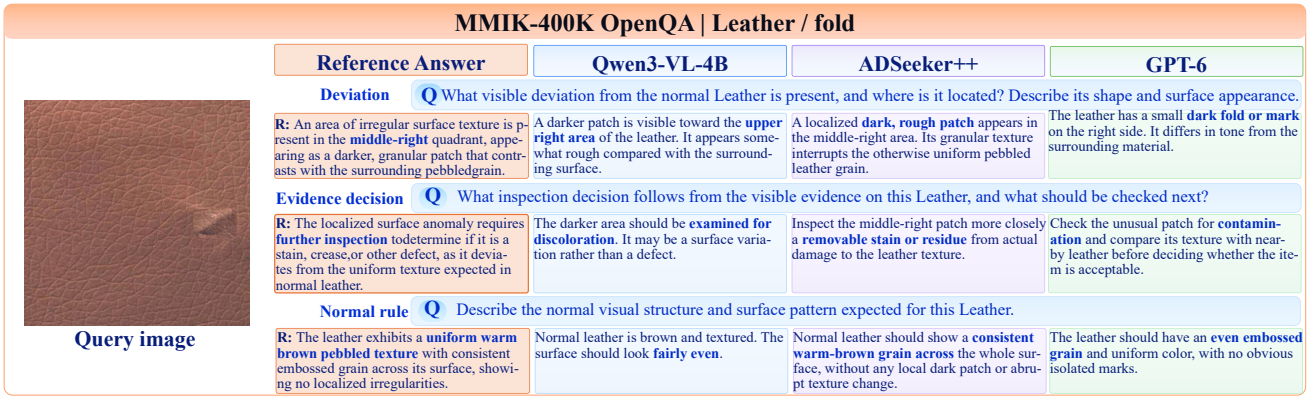

Fig. 8. Qualitative examples of the three **MMIK-400K OpenQA** tasks. Compared with other methods, ADSeeker++ provides more comprehensive responses covering defect locations, appearance characteristics, and supporting evidence in the reference answers, illustrating its potential advantages in open anomaly cognition.

TABLE 6
Seven-task accuracy (%) on **MMIK-400K** ClosedQA.

Method	Anomaly Disc.	Defect Cls./Loc.	Defect Desc./Ana.	Object Cls./Ana.	Avg.
Claude-3.5-Sonnet [58]	53.81	30.46/54.91	64.81/38.76	69.34/66.19	54.04
Gemini-2.5-Flash [59]	62.19	33.85/60.48	62.79/36.69	73.27/68.23	56.79
GPT-5-mini [45]	58.44	35.61/59.77	67.87/39.63	72.14/70.03	57.64
Qwen2.5-VL-7B [56]	60.94	31.54/62.88	59.94/33.87	72.88/65.52	55.37
LLaVA-OneVision-7B [60]	49.88	31.05/51.56	61.73/35.93	71.91/67.05	52.73
InternVL3.5-8B [61]	55.71	32.77/64.83	60.66/35.26	70.62/69.29	55.59
Kimi-VL-A3B-16B [62]	57.23	34.25/61.16	66.96/38.48	72.79/68.56	57.06
MiMo-VL-8B [63]	52.66	35.45/58.71	65.43/37.24	73.38/69.79	56.09
AnomalyR1-7B [12]	63.85	34.18/66.29	61.86/35.71	73.51/68.67	57.72
ADSeeker [16]	55.31	36.37/63.14	68.40/40.12	73.94/69.88	58.15
JUDO-7B [11]	58.98	36.74/64.97	68.72/39.83	73.49/70.21	58.99
AD-Copilot-7B [65]	56.45	34.74/65.62	63.18/37.02	72.28/70.09	57.35
AD-Copilot-Thinking-7B [65]	57.17	36.91/65.15	68.88/40.11	74.06/70.67	58.97
ADSeeker++-4B	**64.97**	**37.08/67.64**	**69.71/40.60**	**74.47/70.81**	**60.76**

5.3.2 *Anomaly Understanding quality analysis*

We evaluate seven structured anomaly understanding tasks on MMAD [14] and the IAC-ClosedQA of MMIK-400K. For MMAD, Qwen3-VL-4B [57] is adopted as the vision-language-model, and the fine-tuning data is constructed following JUDO [11], including MMAD defect samples, Real-IAD [3] images, and corresponding segmentation and domain knowledge QA pairs. During inference, the provided normal templates in MMAD are not used. Instead, Image-RAG retrieves a Top-1 normal reference image from the normal image index, following the standard 1-shot setting. For MMIK-400K, open-source models are first pretrained on SKU for one epoch, while closed-source models acquire industrial knowledge through an RAG knowledge base constructed from one category-level example image and its QA pairs. Unlike MMIK-400k anomaly detection, IAC-ClosedQA uses only the QA-annotated subset, comprising 44,380 images (38,410 normal and 5,970 anomalous) and 310,660 closed-ended QA pairs. Only annotated images are used for ClosedQA training and evaluation. Both splits cover all anomaly categories without QA and image overlap.

Tab. 5 evaluates seven structured anomaly understanding tasks on MMAD. With the fewest parameters, ADSeeker++ achieves the average accuracy of 79.69%, improving 11.16% over the previous ADSeeker [16]. Notably, existing methods typically rely on manually selected normal references from MMAD, while ADSeeker++ automatically retrieves query image-relevant normal references through Image-RAG. Despite removing manual reference selection, ADSeeker++ achieves superior performance, demonstrating the effectiveness of jointly modeling internal normal-anomaly relations and external normal visual knowledge. Specifically, ADSeeker++ achieves 74.50%, 88.43%, 90.99%, and 91.26% on anomaly localization, description, analysis, and object analysis, respectively. It improves anomaly description and object analysis by 3.87% and 3.48% over the best competitors. These results demonstrate that the proposed knowledge enhancement improves fine-grained anomaly understanding with limited parameter overhead.

Tab. 6 further evaluates anomaly understanding of MMIK-400K. Although existing methods acquire certain industrial knowledge through SKU pretraining or category-level examples, their performance remains limited on anomaly classification and analysis, indicating that object-level knowledge alone is insufficient for complex anomaly understanding. By jointly leveraging the internal normal-anomaly relations learned by ADAgent and the external visual references provided by Image-RAG, ADSeeker++ achieves the best performance on all seven tasks, with an average accuracy of 60.76%, improving over JUDO [11] and the previous ADSeeker [16] by 1.77% and 2.61%, respectively. Notably, all methods achieve lower performance on MMIK-400K than on MMAD, demonstrating the higher difficulty of MMIK-400K for anomaly understanding in real industrial scenarios. These results verify the effectiveness of jointly modeling internal relational knowledge and external visual knowledge and the value of MMIK-400k.

5.3.3 *Anomaly Cognition quality analysis*

We further evaluate open anomaly cognition on IAC-OpenQA, including three tasks: Normal Rule, Deviation, and Evidence Decision. Unlike ClosedQA with candidate-based evaluation, OpenQA provides no candidate answers and requires models to autonomously explain anomaly deviations and make detection decisions based on normal pattern knowledge. The industrial knowledge preparation of all models follows the ClosedQA setting. The training set

TABLE 7
MMIK-400K OpenQA results. Each entry reports ROUGE-L F1/ BERTScore F1/ SBERT cosine (%).

Method	Deviation	Evidence Decision	Normal Rule
Claude-3.5-Sonnet [58]	17.21/85.63/41.03	12.92/84.14/36.43	33.81/90.49/73.64
Gemini-2.5-Flash [59]	24.36/88.21/49.73	15.63/85.94/40.68	31.56/89.75/71.88
GPT-5-mini [45]	21.41/87.38/46.27	16.81/86.93/42.47	34.99/90.65/75.44
AnomalyR1-7B [12]	22.82/88.16/49.55	18.05/87.24/43.81	34.07/90.43/74.55
JUDO-7B [11]	23.79/88.35/49.98	18.18/87.54/43.97	34.54/90.76/75.51
ADSeeker [16]	20.50/87.04/46.12	17.46/86.80/42.55	33.75/90.55/74.37
ADSeeker++	**25.26/88.81/51.77**	**18.62/87.85/44.23**	**36.25/90.92/76.28**

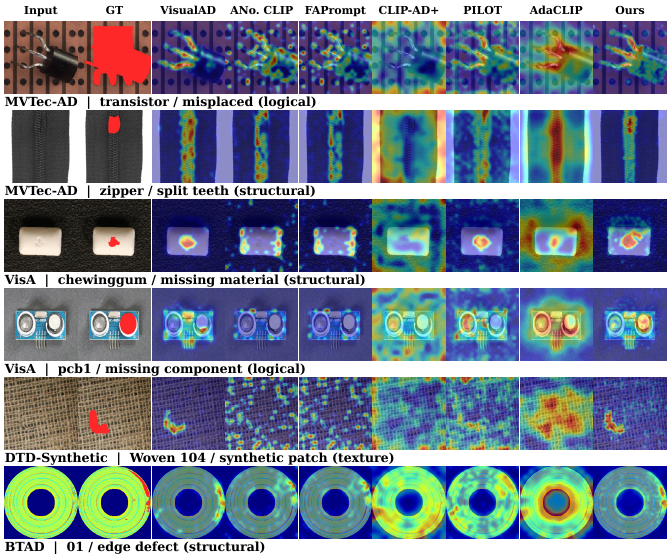

Fig. 9. Qualitative comparison of zero-shot anomaly detection on **MVTec AD, VisA, DTD-Synthetic, and BTAD**, covering logical, structural, and texture anomalies, where Ours denotes ADAgent. Red regions in the GT masks indicate anomalous areas, while warmer colors in the heatmaps represent higher anomaly scores.

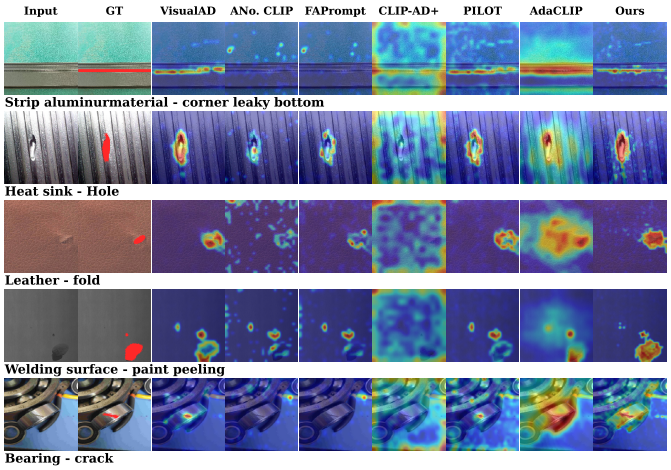

Fig. 10. Qualitative comparison of anomaly detection on **MMIK-400K**.

contains 19,205 normal images, while the test set contains 5,970 anomaly images.

As shown in Tab. 7, ADSeeker++ achieves the best performance on all metrics, demonstrating stronger open anomaly cognition ability. Compared with other methods, ADSeeker++ further incorporates internal normal-anomaly relation modeling and Image-RAG external reference calibration, enabling the model to infer anomaly deviations from normal patterns and generate anomaly explanations. Among the three tasks, Deviation shows the most significant improvement. ADSeeker++ achieves 25.26% ROUGE-L F1 and 51.77% SBERT Cosine, improving over the previous ADSeeker by 4.76% and 5.65%, respectively. This indicates its stronger capability in describing anomaly changes and performing semantic reasoning. Furthermore, ADSeeker++ achieves the best performance on Normal Rule and Evidence Decision, respectively, demonstrating its ability to summarize normal visual rules and perform open judgments. Together with the ClosedQA results, MMIK-400K provides complementary evaluations of structured anomaly understanding and open anomaly cognition. The consistent improvements of ADSeeker++ on both settings demonstrate that jointly modeling internal relational knowledge and external visual knowledge can further advance industrial anomaly cognition beyond anomaly recognition.

5.4 Qualitative Experiments with State-of-the-art

Fig. 8 presents qualitative comparisons of OpenQA anomaly cognition on real-world industrial scenarios of MMIK-400K. Compared with GPT-6 and Qwen3, ADSeeker++ can better understand how anomalies deviate from normal patterns. Existing methods typically generate limited anomaly descriptions, whereas ADSeeker++ provides more complete responses covering anomaly locations, appearance changes, and decision evidence. This advantage comes from the collaboration between internal relational knowledge and external visual knowledge, enabling the model to associate abnormal representations with normal states and further advance anomaly detection toward open anomaly cognition.

Figs. 9 and 10 present qualitative comparisons of anomaly localization across diverse industrial scenarios, covering logical, textural, and structural anomalies. Existing methods are easily affected by normal appearance variations and background information in complex industrial environments, leading to scattered responses or incomplete coverage of anomaly regions. In contrast, ADSeeker++ produces more accurate and complete localization results across different anomaly types. This improvement mainly benefits from ADAgent's explicit modeling of normal-anomaly relations, which enables the model to capture not only abnormal appearance differences but also deviations from normal structures, resulting in more reliable responses for texture changes, local structural, and logical anomalies.

5.5 Ablation Study

5.5.1 *Ablation Experiments of Components*

Tab. 8 shows that jointly using internal and external knowledge outperforms either source alone across all evaluation metrics. Compared with using internal knowledge alone, the joint configuration improves ClosedQA accuracy on MMAD and MMIK-400K by 8.35% and 5.08%, respectively, and increases the OpenQA SBERT score by 6.78%. Internal knowledge provides normal-anomaly relations and anomaly features, while external knowledge supplies category-specific normal visual references. These gains

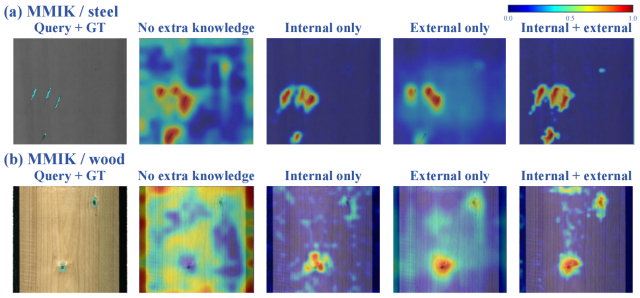

Fig. 11. Visualization of the roles of internal and external knowledge. The joint configuration achieves more complete anomaly coverage while still retaining some background responses.

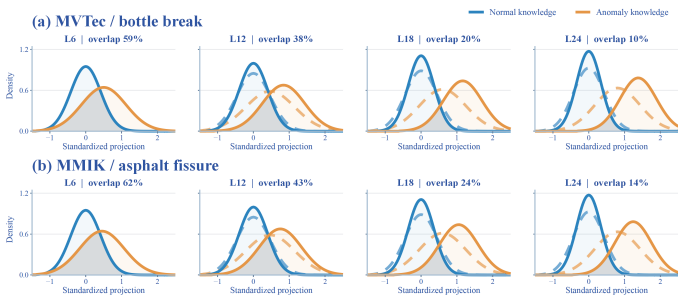

Fig. 12. Visualization of knowledge reinforcement in ADAgent. The figure illustrates the projected distributions of normal and anomaly knowledge, respectively, while dashed and solid curves indicate the distributions before and after knowledge updates. As knowledge is iteratively reinforced, normal and anomaly distributions become more distinguishable.

TABLE 8
Ablation study on ADSeeker++ components on **anomaly understanding and cognition**.

Knowledge Source		ClosedQA		MMIK-400K OpenQA		
Internal	External	MMAD	MMIK-400K	ROUGE-L	BERTScore	SBERT
×	✓	70.06	53.82	21.89	87.33	49.56
✓	×	71.34	55.68	22.38	86.84	50.65
✓	✓	**79.69**	**60.76**	**26.71**	**89.19**	**57.43**

Internal and External denote ADAgent and Image-RAG, respectively. ✓/×: enabled/disabled. ClosedQA reports the mean accuracy over seven tasks. OpenQA reports the average performance over three tasks: Deviation, Evidence Decision, and Normal Rule.

TABLE 9
Ablation study of ADAgent on **zero-shot anomaly detection** (%).

ADAgent Components			MVTec AD		VisA	
RDM	IKR	KGR	I-AUC	P-AUC	I-AUC	P-AUC
✓	×	×	88.6	87.4	81.0	93.6
×	✓	×	88.2	86.8	81.6	91.5
×	×	✓	89.7	88.6	80.8	92.0
×	✓	✓	90.8	89.3	83.5	94.1
✓	×	✓	91.4	89.2	82.3	94.8
✓	✓	×	91.3	90.0	83.4	94.7
✓	✓	✓	**92.9**	**91.3**	**85.0**	**96.4**

RDM: Relation Knowledge Distribution Modeling. IKR: Iterative Knowledge Reinforcement. KGR: Knowledge-Guided Anomaly Reasoning. I/P-AUC denote image-level/pixel-level AUROC.

indicate that the two knowledge sources are complementary for anomaly understanding. In particular, the improvement in semantic agreement on OpenQA suggests that external normal references help the model interpret internal anomaly features.

Fig. 11 illustrates the complementary roles of internal anomaly knowledge and external normal visual knowledge in spatial responses. Without additional knowledge, anomaly responses are widely scattered across the background, making anomaly difficult to distinguish. Internal knowledge captures local anomaly deviations more clearly, while external knowledge provides normal structural references. However, either source alone yields incomplete anomaly coverage. In contrast, their combination highlights multiple anomalous regions and reduces irrelevant background activation. These results suggest that internal knowledge provides anomaly deviation cues, while external knowledge introduces normal-pattern constraints, jointly enabling more complete and focused anomaly responses.

5.5.2 Research on ADAgent

Tabs. 9 and 10 evaluate ADAgent components in cross-dataset zero-shot detection and real-world industrial scenarios, respectively. MMIK-400K covers diverse industrial objects, material textures, and imaging conditions, enabling further evaluation under real production conditions. The full configuration achieves the best image-level and pixel-level AUROC on all three benchmarks. Removing IKR reduces I-AUROC and P-AUROC on VisA by 2.7% and 1.6%, respectively, indicating that IKR help learning normal-anomaly knowledge. On MMIK-400K, removing RDM reduces P-AUROC by 3.6%, demonstrating the contribution of relational knowledge derived from normal distributions and their structural shifts to pixel-level anomaly discrimination in real-world industrial inputs. Removing KGR reduces I-AUROC by 3.1%, indicating that knowledge-guided feature enhancement also improves image-level discrimination. These results support the joint design of ADAgent, and validate its effectiveness in both zero-shot anomaly detection and real-world industrial scenarios.

Tab. 11 evaluates whether ADAgent further supports anomaly understanding and open cognition in vision-language models. Removing IKR reduces ClosedQA accuracy on MMIK-400K by 2.97%, indicating the contribution of knowledge updates to structured anomaly understanding. Removing KGR reduces the OpenQA SBERT score by 3.48%, suggesting that knowledge-guided feature enhancement helps generate responses with better semantic agreement with the reference answers. The full configuration achieves the best results across all metrics, indicating that ADAgent's benefits extend from visual anomaly detection to downstream anomaly understanding and cognition.

Fig. 12 illustrates the knowledge reinforcement of ADAgent. Across deeper layers, the overlap between normal and anomaly knowledge distributions decreases, indicating improved separability. Meanwhile, the anomaly distribution remains broader than the normal distribution to capture diverse anomaly patterns. These observations demonstrate that ADAgent enhances normal-anomaly knowledge separability.

TABLE 10
Ablation study of ADAgent components on **MMIK-400K** ad (%).

ADAgent Components			MMIK-400K	
RDM	IKR	KGR	I-AUROC	P-AUROC
. ✓	×	×	78.9	81.9
×	✓	×	77.3	82.8
×	×	✓	79.5	82.6
×	✓	✓	80.4	84.6
✓	×	✓	81.6	85.3
✓	✓	×	80.6	85.6
✓	✓	✓	**83.7**	**88.2**

TABLE 11
Ablation study of ADAgent for ClosedQA and OpenQA (%).

ADAgent Components			ClosedQA		MMIK-400K OpenQA		
RDM	IKR	KGR	MMAD	MMIK	R-L	B-S	S-C
✓	×	×	76.26	55.43	22.53	87.96	50.58
×	✓	×	75.46	56.36	23.09	87.58	51.72
×	×	✓	77.01	54.56	23.74	88.15	52.57
×	✓	✓	77.67	59.64	25.34	88.49	55.18
✓	×	✓	78.89	57.79	25.03	88.67	56.42
✓	✓	×	78.17	58.95	24.56	88.86	53.95
✓	✓	✓	**79.69**	**60.76**	**26.71**	**89.19**	**57.43**

ClosedQA reports seven-task mean accuracy. R-L/B-S/S-C denote ROUGE-L F1/BERTScore F1/SBERT cosine, averaged over three OpenQA tasks. The intended protocol fixes Qwen SFT and Image-RAG.

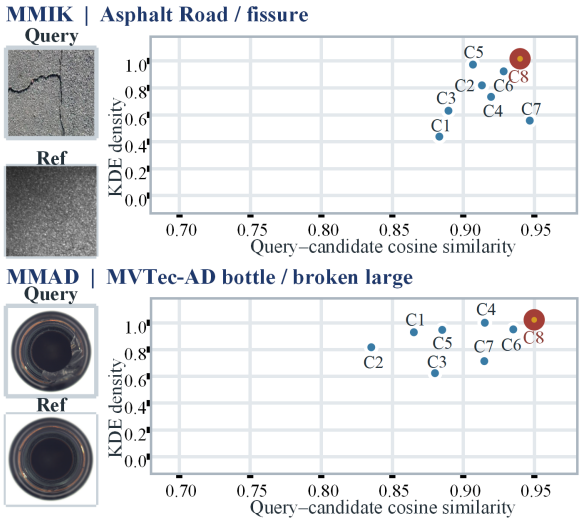

Fig. 13. Distribution modeling in I-RAG. The right panel plots candidate references according to cosine similarity and KDE density, capturing both query similarity and distribution structure. The candidate C8 exhibits high similarity and density, demonstrating how distribution information complements query matching to support more reliable normal reference selection.

5.5.3 Research on Image-RAG

Tab. 12 compares retrieval configurations with normal reference images. Compared with the configuration without GMM or KDE, the full Image-RAG improves ClosedQA accuracy and the OpenQA SBERT score on MMIK-400K by 3.71% and 5.30%, respectively, indicating that similarity-distribution modeling improves the utilization of external normal knowledge. The full configuration also outperforms those using GMM or KDE alone, supporting their complementary roles. Tab. 13 further shows that all evaluated general-purpose and anomaly-specialized models benefit from Image-RAG, suggesting that Image-RAG can enhance visual-language representations across different models.

Fig. 13 illustrates how Image-RAG selects normal visual references. In both examples, candidate C8 exhibits high cosine similarity and KDE density. Specifically, the retrieved references preserve an intact circular rim in the bottle opening and continuous surface textures in the asphalt sample, providing normal-pattern comparisons for identifying structural damage and texture discontinuities, respectively. These results indicate that Image-RAG considers not only appearance similarity but also the normal patterns represented by the retrieved references, supporting the identification and interpretation of anomaly deviations.

TABLE 12
Ablation study of GMM grouping and KDE weighting in Image-RAG on **MMAD and MMIK-400K** (%).

I-RAG Components			ClosedQA		MMIK-400K OpenQA		
Ref.	GMM	KDE	MMAD	MMIK	R-L	B-S	S-C
✓	×	×	76.58	57.05	23.51	87.32	52.13
✓	✓	×	78.48	58.08	25.79	88.02	55.83
✓	×	✓	77.76	59.34	24.74	88.57	54.46
✓	✓	✓	**79.69**	**60.76**	**26.71**	**89.19**	**57.43**

Ref. indicates use of a normal image input.

TABLE 13
Image-RAG ablation study on **MMAD and MMIK-400K** (%).

Model	**I-RAG**	ClosedQA		MMIK-400K OpenQA		
		MMAD	MMIK	R-L	B-S	S-C
General-purpose VLMs						
Gemini-2.5-Flash [59]	×	80.08	56.79	23.85	87.97	54.10
	✓	82.01	59.56	25.19	88.49	56.53
GPT-5-mini [45]	×	77.66	57.64	24.40	88.32	54.73
	✓	79.38	60.03	25.74	88.81	57.14
Anomaly-specialized VLMs						
AnomalyR1-7B [12]	×	77.30	57.72	24.98	88.61	55.97
	✓	79.82	60.48	26.03	89.27	58.01
JUDO-7B [11]	×	81.20	58.99	25.50	88.88	56.49
	✓	83.06	61.27	26.66	89.31	58.76
ADSeeker++-4B	×	71.34	55.68	22.38	86.84	50.65
	✓	79.69	60.76	26.71	89.19	57.43

5.5.4 Research on MMIK-400K

Tab. 14 compares direct IAC fine-tuning with SKU knowledge pretraining and IAC fine-tuning. Adding SKU pretraining improves ClosedQA accuracy by 1.55% and the OpenQA SBERT score by 1.67%, with improvements in the remaining OpenQA metrics as well. The consistent gains across both response formats indicate that knowledge acquired through SKU pretraining supports subsequent anomaly cognition tasks, supporting the design of

TABLE 14
Comparison of SKU scene-knowledge pretraining before IAC fine-tuning **MMIK-400K** (%).

		ClosedQA	MMIK-400K OpenQA		
SKU	IAC	MMIK	R-L	B-S	S-C
×	✓	59.21	25.54	88.94	55.76
✓	✓	60.76	26.71	89.19	57.43

MMIK-400K that connects industrial knowledge learning with anomaly cognition evaluation.

6 Conclusion

The diversity of industrial objects, and complex imaging conditions in real-world scenarios pose significant challenges to anomaly understanding and cognition in Large Vision-Language Models (LVLMs). To address the limited category coverage of existing benchmarks and the fragmentation of knowledge learning, anomaly understanding, and cognition tasks, we construct MMIK-400K, a large-scale multi-modal industrial knowledge benchmark. To the best of our knowledge, MMIK-400K is the first and largest-scale industrial multi-modal benchmark to unify Scene Knowledge Pretraining, Anomaly Understanding, OpenQA Cognition Evaluation. MMIK comprises 461,388 real industrial images across 432 object categories, together with 1,271,155 closed and 133,140 open QA pairs, providing a unified foundation for knowledge learning and anomaly cognition evaluation in industrial LVLMs. Based on MMIK-400K, we further propose ADSeeker++, a framework that jointly models internal and external knowledge. ADAgent learns internal anomaly knowledge by modeling normal visual distributions and normal-to-anomaly structural relations. Image-RAG employs distribution modeling to select query-relevant normal visual references, providing external normal-pattern knowledge. Extensive experiments demonstrate the effectiveness of ADSeeker++ in anomaly detection, structured anomaly understanding, and open anomaly cognition. The results further highlight the complementary roles of internal relational knowledge and external normal visual knowledge in improving industrial anomaly understanding and cognition. We expect MMIK-400K and ADSeeker++ to facilitate future research on cross-scenario knowledge transfer, retrieval-augmented anomaly reasoning, and reliable anomaly cognition in real-world industrial environments.

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